Lagrange's Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of lagrange's theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

MA-01 / Modern Algebra / Report v1

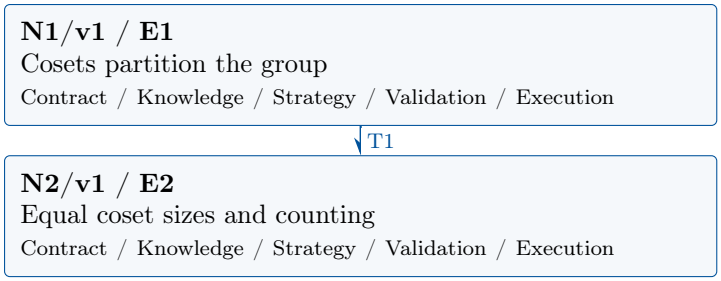
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a finite group G and subgroup $H \leq G$, prove $|G| = [G : H]|H|$, hence $|H|$ divides $|G|$.*

A. Foundations and permitted imports

G is finite and H is a subgroup. Left cosets are sets $gH = \{gh : h \in H\}$; they are not being endowed with a quotient group structure. Use group inverses, bijections, equivalence classes, and finite counting.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The left cosets give a disjoint partition of G.

Receiving activity. N2: Equal coset sizes and counting

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Partition and left multiplication; prior exports E1/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Cosets partition the group

Object and role. Characterize when two cosets intersect.

Inputs. Finite G and subgroup H; foundational imports stated in §1

C — Contract

Own E1/v1: The left cosets give a disjoint partition of G.

K — Knowledge

Retain Finite G and subgroup H; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Characterize when two cosets intersect.

V — Validation

No normality is needed for this set partition.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Every g belongs to gH since $1 \in H$.

N1.2 If $gH \cap kH$ contains $x = gh_1 = kh_2$, then $k^{-1}g = h_2h_1^{-1} \in H$.

N1.3 Therefore every gh equals $k(h_2h_1^{-1}h) \in kH$, and conversely, so the two cosets are equal.

N1.4 Thus distinct cosets are disjoint and all cosets partitionG.

Export E1/v1. The left cosets give a disjoint partition of G.

Acceptance gate. No normality is needed for this set partition. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: No normality is needed for this set partition. Recheck dependent nodes N2.

B. N2/v1: Equal coset sizes and counting

Object and role. Supply a bijection for each coset.

Inputs. Partition and left multiplication; prior exports E1/v1

C — Contract

Own E2/v1: Lagrange's formula and divisibility.

K — Knowledge

Retain Partition and left multiplication; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Supply a bijection for each coset.

V — Validation

Divisibility is a necessary condition for subgroup order, not a universal existence theorem.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 For any g the map $H \to gH$, $h \mapsto gh$, is surjective by definition and injective by left cancellation.

N2.2 Every coset has|H| elements.

N2.3 If there are[G:H] cosets, finite disjoint counting $gives|G| = [G : H]|H|$.

N2.4 Thus|H| divides|G|.

N2.5 The cases $H = \{1\}$ and $H = G$ are included.

Export E2/v1. Lagrange's formula and divisibility.

Acceptance gate. Divisibility is a necessary condition for subgroup order, not a universal existence theorem. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Divisibility is a necessary condition for subgroup order, not a universal existence theorem. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Cosets partition the group

1. Every g belongs to gH since $1 \in H$.
2. If $gH \cap kH$ contains $x = gh_1 = kh_2$, then $k^{-1}g = h_2h_1^{-1} \in H$.
3. Therefore every gh equals $k(h_2h_1^{-1}h) \in kH$, and conversely, so the two cosets are equal.
4. Thus distinct cosets are disjoint and all cosets partitionG.

B. §4.N2 — Equal coset sizes and counting

1. For any g the map $H \to gH$, $h \mapsto gh$, is surjective by definition and injective by left cancellation.
2. Every coset has|H| elements.
3. If there are[G:H] cosets, finite disjoint counting $gives|G| = [G : H]|H|$.
4. Thus|H| divides|G|.
5. The cases $H = \{1\}$ and $H = G$ are included.

□

V. WORKED ORCHESTRATION

In S_3, take $H = \{1, (12)\}$, of order 2. Its left cosets are H,(13)H,(23)H, each with two elements, giving $6 = 3 \cdot 2$. H is not normal, which confirms that normality is unnecessary for Lagrange's counting argument.

A. Boundary and hypothesis audit

A proposed quotient-group multiplication on these cosets requires normality, which thisH lacks. Validation prevents a counting export from being used as a quotient-construction certificate; the partition remains valid.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A proposed quotient-group multiplication on these cosets requires normality, which thisH lacks. Validation prevents a counting export from being used as a quotient-construction certificate; the partition remains valid.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.